

Supply Chain Analysis Report

Prepared from Python-based exploratory analysis and predictive modeling

Executive Summary

1. Business Problem

A global e-commerce organization operating across multiple regions faces inconsistent order fulfillment performance. Actual shipping times often deviate from scheduled timelines, creating late deliveries, customer dissatisfaction, and profit volatility.

The business objective is to identify operational bottlenecks, quantify the financial effect of delay, and build a predictive system that helps reduce delays and improve fulfillment efficiency.

2. Data Overview and Preparation

The original dataset contained 180,519 rows and 53 columns. After dropping non-essential identifiers, personal data, and unrelated fields, and removing cancelled shipping records, the cleaned analysis dataset contained 172,765 rows and 20 columns.

Key preparation steps included:

- Removing personal, address, and product image fields that are not useful for operational analytics.
- Filtering out 'Shipping canceled' records to focus on completed fulfillment flow.
- Converting order and shipping timestamps to datetime format.
- Engineering delay features: Order Processing Time, Delay, Is_Delayed, order_month, order_day, and order_hour.
- Creating a Profitability Flag to classify orders as Profit, Loss, or Break-even.

3. Key Business Metrics

Metric	Value
Total orders analyzed	172,765
Late deliveries	94,523
Late delivery rate	54.71%
On-time delivery rate	45.29%
90th percentile delay	3 days
Total profit	7.5M \$
Estimated loss due to delays	2.1M \$
Profitability distribution	Profit 80.661% Loss 18.693% Break-even 0.646%

4. Exploratory Data Analysis Findings

The dataset shows a strong imbalance toward delayed shipments. In the cleaned data, 94,523 orders were flagged as delayed, compared with 78,242 on-time orders. Profitability is also uneven: 80.661% of orders were profitable, 18.693% generated a loss, and only 0.646% were break-even.

The delay distribution and profitability analysis indicate that delay is not a minor exception; it is a core operational issue that materially affects financial outcomes.

Observed patterns from the visual analysis:

- Delay frequency varies by delay day, confirming that small delivery slips accumulate across the order base.
- Profit changes with delay behavior, showing that late orders are associated with weaker profitability outcomes.
- Certain categorical groups such as region, customer segment, shipping mode, order status, type, and department name display different late-delivery rates.

5. Bottleneck Detection and Root Cause Analysis

To identify operational bottlenecks, the notebook evaluates delay percentages across multiple business dimensions. The analysis compares late-delivery rates by order region, customer segment, shipping mode, order status, product type, and department name.

A regional deep-dive was also performed to identify the strongest drivers of late delivery in a selected region. The approach ranks factor levels by their delay percentage, helping operations teams focus on the most problematic combinations of channel, segment, and order condition.

The key analytical value of this stage is prioritization: it points the business toward the categories most associated with delay instead of treating all orders equally.

6. Time-Based Analysis

The temporal analysis examines delay behavior by month, day of week, and hour of order placement. The order timeline in the dataset spans from 2015-01-03 to 2018-02-06, which supports a multi-year operational view.

- Monthly trends identify whether delivery performance deteriorates or improves during specific periods of the year.
- Day-of-week patterns reveal whether workload concentration or logistics cadence affects punctuality.
- Hourly analysis highlights whether the time an order is created has a measurable relationship with delay risk.

7. Predictive Modeling

A Random Forest classifier was used to predict late delivery risk. The feature set included product type, scheduled shipment days, category name, customer segment, department name, order region, shipping mode, and engineered time features.

Categorical variables were frequency-encoded to reduce dimensionality. Because the target variable was imbalanced, SMOTE was applied to the training set, resulting in perfectly balanced classes before model fitting.

Class balance before and after SMOTE:

- Before balancing: Counter({1: 79182, 0: 59030})
- After balancing: Counter({0: 79182, 1: 79182})

Test-set performance of the Random Forest model was as follows: accuracy 0.74, precision 0.79, and recall 0.75. The classification report showed class-0 F1 of 0.70 and class-1 F1 of 0.77, with weighted average F1 approximately 0.74.

8. Business Implications

- Late delivery is widespread enough to require operational intervention rather than isolated exception handling.
- Delay is associated with measurable financial impact and should be managed as a profit-protection lever, not only a service metric.
- The model's predictive power is sufficient for early-warning workflows, exception prioritization, and routing to operational teams.
- Regional and categorical differences suggest that a one-size-fits-all logistics policy is not optimal.

9. Recommendations

- Prioritize high-risk shipping modes, regions, and customer segments in daily operational monitoring.
- Introduce alerting for orders whose predicted delay risk exceeds a defined threshold.
- Investigate root-cause clusters where delay percentage and average delay are both elevated.
- Reassess cut-off times, dispatch planning, and carrier assignment rules for periods showing higher delay percentages.
- Track delay-related profit leakage as a recurring KPI alongside on-time delivery rate.

10. Conclusion

This project demonstrates that delivery delays are both measurable and predictable. The analysis identifies delay concentration, financial impact, and segment-level variation, while the model provides a practical starting point for early risk detection. For an MNC environment, the highest value comes from combining descriptive analytics with predictive operational controls so that delay risk is reduced before it affects customers and profit.